

Five Days at Fremont — Part D

AC101 integrated Tesla case · worked in Lecture 4

*Part D of the week-long AC101 Tesla case — worked in Lecture 4. The full fact-vs-fiction note and company background are in the Part A handout (the complete case is on the course site). In short: Tesla is real and the filed figures are real; **every product-costing scenario, internal figure and decision here is hypothetical**; exhibits are labelled REAL, SIMULATED or ESTIMATE. Every question is worked and discussed in the lecture.*

Part D — Thursday: make or buy — the battery question

Thursday's diary says "board pre-read", and the pre-read contains a memo from a colleague that is going to teach you the most important single idea of the week — by being wrong.

The board question: should Tesla make battery cells in-house, or buy them from Panasonic? The memo's answer:

"Our in-house full cost is \$70 variable plus \$800m of annual fixed spread over 87.7m kWh — call it \$79/kWh. Panasonic lands at \$90. Make: we save \$11."

It sounds fine. The arithmetic is even correct. And the reasoning can be exactly backwards, because the memo never asked *which costs are allowed into the comparison*.

The rule that sorts it out has two gates. A cost is **relevant** to a decision only if it is (1) **future** — not yet incurred, still changeable — and (2) **different between the alternatives**. Fail either gate and the cost is noise for *this* decision, however large and however painful.

Gate 1 dispatches the biggest number on the table. Tesla reported **\$6.2bn invested in Gigafactory Nevada by early 2023, with a \$3.6bn expansion underway** (real figures, labelled in Exhibit D-1). All of it is **sunk** — gone whichever way the cell decision goes — so none of it may touch the choice. What *can* matter about the factory is only what is still changeable: its resale value, or what else the floor could earn.

Gate 2 dispatches the memo's \$800m. For the decision actually on the table — *where should the next batch of cells come from, on the existing line* — the \$800m of ongoing fixed cost is there whether the next batch is made or bought. It does not differ, so it does not count. The relevant comparison is **\$70 versus \$90 per kWh: making saves \$20**. The memo's conclusion (make) happens to survive; its reasoning (full cost) does not — and Question 3 shows how quickly that luck runs out, because the moment extra demand needs a **new** line at \$400m a year, that fixed cost is future-and-different, and making only wins beyond a break-even volume of extra demand.

Midday brings an interruption that runs on the same rule. A rental company offers to buy **50,000 Model 3s at \$30,000 each** — \$9,000 below list (Exhibit D-2, illustrative). Full cost per car is \$33,000, so the offer "loses \$3,000 a car" ... on a full cost that includes \$8,000 of allocated fixed overhead that exists either way. With spare capacity, the relevant comparison is price against *variable* cost, and the order adds a quarter of a billion dollars of contribution. At *full* capacity the logic flips: each special car displaces a full-price sale, and the displaced contribution — invisible on any invoice — is a real **opportunity cost**. Sunk costs are visible but irrelevant; opportunity costs are invisible but relevant. Instinct weights them exactly backwards.

Then, late afternoon, the board widens the question — and the same \$800m changes sides. Not "where does the next batch come from?" but **"should Tesla be in the cell business at all, or exit entirely and clear the floor?"** Against *that* alternative, the \$800m of ongoing cash fixed cost is avoidable — closing the line stops it — so it now passes the "differs" gate. And the cleared floor could earn roughly **\$1,100m a year of**

Megapack contribution (profit forgone, not revenue – Exhibit D-1, illustrative), which is an opportunity cost of staying. Same dollars, different question, different relevance – and, as Question 5 shows, a different answer.

The week's quietest lesson lands here: the numbers narrow the choice; judgement finishes it. The real Tesla both makes cells (its 4680 programme) *and* buys them (Panasonic, LG, CATL) – because supply security and proprietary battery technology are worth real money that appears in no relevant-cost table.

Assignment questions – Part D

1. State the relevant-cost principle (future **and** differs). Apply gate 1 to the Gigafactory's ~\$6.2bn invested to date: is it relevant to make-versus-buy? What, if anything, about the factory can still be relevant?
2. Your colleague's memo argues: "full cost in-house = \$70 + \$800m ÷ 87.7m kWh ≈ \$79/kWh, versus \$90 from Panasonic – make." For the decision actually on the table – where the next batch of cells comes from, on the existing line – rebuild the comparison with relevant costs only, and explain which gate the \$800m fails and why the conclusion happens to survive while the reasoning does not.
3. Extra demand beyond current capacity would need a new \$400m-a-year line. Find the break-even extra volume and state the decision rule either side of it.
4. The special order (Exhibit D-2): show that the full-cost view rejects it and the relevant view accepts it with spare capacity. Then the hinge: at full capacity, what does each special car displace? Give the decision and the minimum acceptable price in each capacity state.
5. ► The board widens the question: exit the battery business entirely and clear the floor. Show that the \$800m now passes the "differs" test and that the forgone Megapack contribution is a relevant opportunity cost, and compute the relevant totals of MAKE versus BUY. Two sentences on why the same \$800m changed status – and two qualitative factors that explain why the real Tesla does both.
6. Your colleague, challenged on the \$79 memo, produces the fuller worksheet in Exhibit D-1 (further lines). For the decision in Question 2 – where the next batch of cells comes from, on the existing line – go through every figure now on the worksheet and sort it into "counts" or "ignore for this decision". For each figure you ignore, name which test it fails: not future, or does not differ. State the one assumption about the existing line that lets you set the \$800m and the depreciation aside. Then work out what the padded "full cost" per kWh would have told you, and explain in two sentences why the relevant comparison disagrees.
7. ► Question 4 gave you two clean cases: all spare capacity, or none. Reality is usually in between. Take Exhibit D-2 with the Question 7 note – the Model 3 line can build 500,000 cars a year and currently sells 470,000, and the rental company still wants 50,000 cars at \$30,000. Work out how many of the 50,000 use idle capacity and how many displace full-price sales, and hence the relevant profit or loss of accepting the whole order. Give the minimum price at which you would take the whole order. Then find how much spare capacity the line would need for the order to be worth taking at \$30,000, and state the assumption your answer depends on.

Exhibits – Part D

Exhibit D-1 – Battery cell cost data (make versus buy) SIMULATED – benchmarked on the BloombergNEF battery cost index and industry reports. **Scope flag: these are cell-level variable costs – do not compare them to ~\$115–140/kWh pack-level prices (BloombergNEF 2024), which add pack assembly, housing and battery management; a different scope.**

Item	\$/kWh
In-house cell, variable cost	70
– raw materials	45
– direct labour	8
– manufacturing overhead (variable)	12
– quality control	3
– logistics	2
Panasonic, landed	90

Item	\$/kWh
– quoted price	85
– logistics	5

Other data	Figure
In-house ongoing cash fixed cost (utilities, maintenance, salaried staff, leases) – avoidable only if the line closes; excludes the sunk build	\$800m/yr
Annual cell demand (the 87.67m kWh battery driver total from Exhibit C-2; 87.7m in round figures)	87.7m kWh

Further lines from the colleague's worksheet (SIMULATED – illustrative; the fuller worksheet the colleague produces when the \$79 memo is challenged. Not all of these belong in the make-or-buy comparison.)

Item	\$/kWh
Corporate head-office cost allocated to the cell line (spread by kWh)	6
Depreciation of the existing cell line (accounting spread of the sunk Gigafactory build)	9
Panasonic's list quote before Tesla's negotiated volume discount	95

*Context (REAL): Tesla reported \$6.2bn invested in Gigafactory Nevada by early 2023, with a \$3.6bn expansion underway. Illustrative additions (SIMULATED): a new production line would cost \$400m/yr in fixed costs; the freed floor could earn ~\$1,100m/yr of Megapack **contribution** (profit forgone, not revenue).*

Exhibit D-2 – The rental-fleet special order SIMULATED – labelled illustrative; simplified Model 3 costs.

Item	Figure
Offer	50,000 Model 3s at \$30,000 per car
List price	\$39,000
Variable cost per car	\$25,000
Allocated fixed overhead per car	\$8,000
"Full cost" per car	\$33,000

For Question 7 (SIMULATED – illustrative): the Model 3 line is built for its planned 500,000 cars a year (Exhibit C-1) and is currently running at 470,000. The rental order would be additional to current sales.